Object-Oriented and Classical Software Engineering

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CHAPTER 6

TESTING

Testing

- Testing is continuous – it is not sufficient to test the product of the workflow only at the end of the workflow

- There are two basic types of testing
  - Execution-based testing
    - Testing through test cases
  - Non-execution-based testing
    - Testing software without running test cases

Testing (contd)

- "V & V"
  - *Verification*
    - Determine if the workflow was completed correctly
    - Takes place at the end of each workflow
  - *Validation*
    - Determine if the product as a whole satisfies its requirements
    - Takes place just before the product is delivered to the client

6.1.1 Software Quality Assurance

- The members of the SQA group must ensure that the developers are doing high-quality work
  - At the end of each workflow
  - When the product is complete
- In addition, quality assurance must be applied to
  - The process itself
    - Example: Standards to which the software must conform

6.1.2 Managerial Independence

- There must be managerial independence between
  - The development group
  - The SQA group
- Each group must fall under their own manager
- Neither manager should be able to overrule the other

Managerial Independence (contd)

- More senior manager must decide whether to
  - Deliver the product on time but with faults, or
  - Test further and deliver the product late
- The decision must take into account the interests of the client and the development organization
- The additional cost of having a SQA is relatively small compared to the resulting benefit – higher quality software

6.2 Non-Execution-Based Testing

- Testing software without running test cases

- Examples
  - Reviewing software
    - Walkthroughs
    - Inspections
  - Analyzing software mathematically

- Underlying principle
  - We should not review our own work – “blind spots”

6.2.1 Walkthroughs

- A walkthrough team consists of from four to six members
- It includes at least one representatives from
  - The team responsible for the current workflow
  - The team responsible for the next workflow
  - A client representative
  - The SQA group (should chair the walkthrough)
- The walkthrough is preceded by preparation
  - Each reviewer should draw up two lists
    - Items not understood
    - Items that reviewer believes are incorrect

6.2.2 Managing Walkthroughs

- The walkthrough team is chaired by the SQA representative because the SQA has the most to lose if the walkthrough is performed poorly and faults slip through
- Person leading walkthrough guides other members through the document to uncover faults
- In a walkthrough we detect faults, not correct them
  - A correction produced by a committee is likely to be of low quality
  - The cost of a committee correction is too high
  - Not all items flagged are actually incorrect
  - A walkthrough should not last longer than 2 hours, there is no time to correct faults as well

Managing Walkthroughs (contd)

- A walkthrough can be participant-driven or document-driven
- Participant-driven
  - Participants present their lists of unclear items and items they think are incorrect
  - Representative of team must respond to each query
    - Clarifying what is unclear to the reviewer
    - Either agreeing that there is a fault or explaining why the reviewer is mistaken

Managing Walkthroughs (contd)

- Document-driven
  - A person responsible for the document walks the reviewers through the document
  - Reviewers interrupt with their prepared comments or comments triggered by the presentation
- Document-driven is
  - More thorough
  - Leads to the detection of more faults, since the majority is found spontaneously by the presenter - Verbalization leads to fault finding

6.2.3 Inspections

- An inspection has five formal steps
  - Overview
    - Overview of document is given by individuals responsible for document
  - Preparation
    - Participants try to understand document in detail
  - Inspection
    - One participant walks through the document with the inspection team (All branches must be followed) and the fault finding commences (faults are not corrected)
    - Moderator produce a report
  - Rework
    - Faults and problems noted in the written report are resolved
  - Follow-up
    - Moderator ensures that every issue raised has been resolved

Statistics on Inspections

- Warning
- Fault statistics should never be used for performance appraisal
  - “Killing the goose that lays the golden eggs”

6.2.4 Comparison of Inspections and Walkthroughs

- Walkthrough
  - Two-step process
    - Preparation
    - Analysis
- Inspection
  - Five-step, formal process
    - Overview
    - Preparation
    - Inspection
    - Rework
    - Follow-up
- Inspection takes much longer than a walkthrough

6.3 Execution-Based Testing

- Organizations spend up to 50% of their software budget on testing
  - But delivered software is frequently unreliable

- Dijkstra
  - “Program testing can be a very effective way to show the presence of bugs, but it is hopelessly inadequate for showing their absence”

- The following sections are not for test and exam purposes but must be read:
  - Section 6.5
  - Section 6.6

6.7 When Testing Stops

- Only when the product has been irrevocably discarded